

EE 4113 Homework 1 Assignment

Homework 1 Overview:

This assignment is worth 20 points. All *Homework 1* questions are based on the content available through the **Week 1 – Complex Sinusoids** module and the **Week 2 – Impedance** module listed in the EE 4113 Blackboard shell under **Lecture Slides**.

Homework 1 Questions:1. Properties of a Sinusoidal Function Expressed in the Time-Domain

Given a periodic signal described by the function,

$$v(t) = 40 \cos(100\pi t + 60^\circ) \text{ V}$$

- (1 point) What is the maximum amplitude of the voltage?
- (1 point) What is the frequency of the signal in Hertz?
- (1 point) What is the frequency of the signal in radians per second?
- (1 point) Identify the phase angle (in degrees).
- (1 point) What is the phase angle in radians?
- (1 point) What is the period in milliseconds?

2. Transformation Between Time-Domain and Frequency Domain Representations

Find the phasor transform for the following trigonometric functions (include SI units in your answer).

- (1 point) $v(t) = 170 \cos(377t - 40^\circ) \text{ V}$
- (1 point) $i(t) = 10 \sin(1000t - 20^\circ) \text{ A}$
- (1 point) $i(t) = [5 \cos(\omega t - 36.87^\circ) + 10 \cos(\omega t - 53.13^\circ)] \text{ A}$

3. Phasor Operations

Given the functions $f(t)$ and $g(t)$,

$$f(t) = 12 \cos(200t + 45^\circ)$$

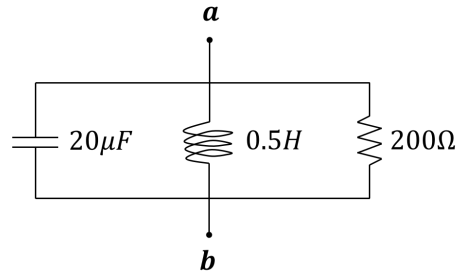
$$g(t) = 5 \sin(200t + 102^\circ)$$

Solve the following operations using phasor notation and/or complex sinusoids,

- (1 point) $h_a(t) = f(t) + g(t)$
- (1 point) $h_b(t) = f(t) \cdot g(t)$
- (1 point) $h_c(t) = f(t) / g(t)$

4. Circuit Simplifications Using Impedance

(4 points) Given the following circuit compute the equivalent impedance between points **a** and **b**, assume we are operating at $\omega = 60 \text{ rad/sec}$.

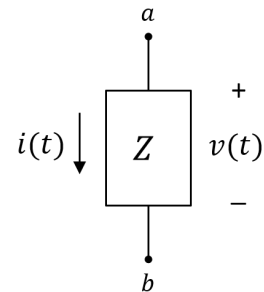


5. Voltage, Current and Power Calculations in Circuit Analysis

For the following diagram, assume that the voltage $v(t)$ and impedance Z are given as

$$v(t) = 12 \cos(\omega t + 20^\circ)$$

$$Z = -1.6971 + j1.6971$$



- (3 points) Compute the current $i(t)$.
- (1 point) Taking the voltage $v(t)$ and the calculated current $i(t)$ from part 'a' of this problem, compute the real power P , reactive power Q , and apparent power $|S|$.

1. Properties of a Sinusoidal Function Expressed in the Time-Domain

Given a periodic signal described by the function,

$$v(t) = 40 \cos(100\pi t + 60^\circ) \text{ V}$$

- (1 point) What is the maximum amplitude of the voltage?
- (1 point) What is the frequency of the signal in Hertz?
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- (1 point) What is the phase angle in radians?
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a) maximum amplitude $A = 40$

b) normal frequency $f = \frac{1}{T}$ $\omega = 2\pi f = \frac{2\pi}{T}$ $f = \frac{100\pi}{2\pi} = 50 \text{ Hz}$
 $T = \frac{1}{f} = \frac{1}{50}$ $f = \frac{\omega}{2\pi}$

c) radian frequency $\omega = 2\pi f = \frac{2\pi}{T}$, $f = \frac{1}{T} \Leftrightarrow T = \frac{1}{f} = \frac{1}{50} = 0.02 \text{ s}$

$$\omega = \frac{2\pi}{0.02 \text{ s}} = 314.1592 \approx 314.15^\circ$$

d) $40 \cos(100\pi t + 60^\circ) \text{ V}$ ϕ in degrees?

$$\phi = 60^\circ$$

e) ϕ in radians

$$60^\circ \times \frac{\pi}{180^\circ} = \frac{60^\circ}{180^\circ} \pi = \frac{1}{3} \pi = \frac{\pi}{3}$$

f) Period in ms?

$$T = 0.02 \text{ s} \quad 20 \text{ ms} \quad T = 20 \text{ ms}$$

2. Transformation Between Time-Domain and Frequency Domain Representations

Find the phasor transform for the following trigonometric functions (include SI units in your answer).

- (1 point) $v(t) = 170 \cos(377t - 40^\circ) \text{ V}$
- (1 point) $i(t) = 10 \sin(1000t - 20^\circ) \text{ A}$
- (1 point) $i(t) = [5 \cos(\omega t - 36.87^\circ) + 10 \cos(\omega t - 53.13^\circ)] \text{ A}$

a) $A = Ae^{j\phi} = A \cos \phi + jA \sin \phi$

$$V(t) = 170 \cos(377t - 40^\circ) \text{ V}$$

$$\frac{170 \cos(-40) + j170 \sin(-40) \text{ V}}{\text{or } |A| \angle \phi} \quad \boxed{170 \angle -40^\circ \text{ V}}$$

b) $i(t) = 10 \sin(1000t - 20^\circ) \text{ A}$ $\cos(\theta) = \sin(\theta - 90^\circ)$

$$|A| = 10 \rightarrow \boxed{10 \angle -110^\circ \text{ A}}$$

c) $i(t) = [5 \cos(\omega t - 36.87^\circ) + 10 \cos(\omega t - 53.13^\circ)] \text{ A}$

$$5 \cos(\omega t - 36.87^\circ) \text{ A}$$

$$|A| \angle \phi$$

$$5 \angle -36.87^\circ \text{ A}$$

$$10 \cos(\omega t - 53.13^\circ) \text{ A}$$

$$|A| \angle \phi$$

$$10 \angle -53.13^\circ \text{ A}$$

$$\boxed{5 \angle -36.87^\circ \text{ A} + 10 \angle -53.13^\circ \text{ A}}$$

3. Phasor Operations

Given the functions $f(t)$ and $g(t)$,

$$f(t) = 12 \cos(200t + 45^\circ)$$

$$g(t) = 5 \sin(200t + 102^\circ)$$

Solve the following operations using phasor notation and/or complex sinusoids,

- (1 point) $h_a(t) = f(t) + g(t)$
- (1 point) $h_b(t) = f(t) \cdot g(t)$
- (1 point) $h_c(t) = \frac{f(t)}{g(t)}$

$$h_a(t) = (6\sqrt{2} + j4.89) + j(6\sqrt{2} + j4.89) = \boxed{13.37 + j13.37}$$

$$h_b(t) = (12 \angle 45^\circ) (5 \angle 12^\circ) = \boxed{60 \angle 57^\circ}$$

$$h_c(t) = \frac{12 \angle 45^\circ}{5 \angle 12^\circ} = \frac{12}{5} \angle 45^\circ - 12^\circ = \boxed{2.4 \angle 33^\circ}$$

$$f(t) + g(t) \quad f(t) = 12 \cos(200t + 45^\circ)$$

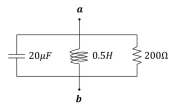
$$f(t) = A \cos \phi + jA \sin \phi = 12 \cos(45^\circ) + j12 \sin(45^\circ) = 12 \cos\left(\frac{\pi}{4}\right) + j12 \sin\left(\frac{\pi}{4}\right) = 12 \left(\frac{\sqrt{2}}{2}\right) + j12 \left(\frac{\sqrt{2}}{2}\right) = 6\sqrt{2} + j6\sqrt{2}$$

$$g(t) = 5 \sin(200t + 102^\circ) \Rightarrow 5 \cos(200t + 12^\circ) \quad 12^\circ \cdot \frac{\pi}{180^\circ} = \frac{\pi}{15}$$

$$A \cos \phi + jA \sin \phi = 5 \cos(12^\circ) + j5 \sin(12^\circ) = 5 \cos\left(\frac{\pi}{15}\right) + j5 \sin\left(\frac{\pi}{15}\right) = 4.89 + j0.89$$

4. Circuit Simplifications Using Impedance

(4 points) Given the following circuit compute the equivalent impedance between points a and b , assume we are operating at $\omega = 60 \text{ rad/sec}$.



$$V = I \cdot Z$$

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$$Z = \frac{V}{I}$$

$$Z_R = R = 200 \Omega$$

$$Z_L = j\omega L = j(60)(0.5) = j30$$

$$Z_C = j\left(\frac{-1}{\omega C}\right) = j\left(\frac{-1}{(60)(20 \cdot 10^{-6})}\right) = -j833.33 \Omega$$

$$Z_{eq} = Z_R + Z_L + Z_C \text{ (Series)}$$

$$\rightarrow \text{Parallel } \frac{1}{Z_{eq}} = \frac{1}{Z_R} + \frac{1}{Z_L} + \frac{1}{Z_C}$$

$$\rightarrow Z_{eq} = \left(\frac{1}{Z_R} + \frac{1}{Z_L} + \frac{1}{Z_C}\right)^{-1} \Rightarrow Z_{eq} = \left(\frac{1}{200 \Omega} + \frac{1}{j30 \Omega} + \frac{1}{-j833.33 \Omega}\right)^{-1}$$

$$= 200 \Omega + j30 \Omega - j833.33 \Omega$$

$$= 200 \Omega - j803.33$$

$$Z_{eq} = \sqrt{(200)^2 + (-803.33)^2} \angle \tan^{-1}\left(\frac{-803.33}{200}\right)$$

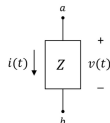
$$= 827.85 \angle -76.01^\circ$$

5. Voltage, Current and Power Calculations in Circuit Analysis

For the following diagram, assume that the voltage $v(t)$ and impedance Z are given as

$$v(t) = 12 \cos(\omega t + 20^\circ)$$

$$Z = -1.6971 + j1.6971$$



a. (3 points) Compute the current $i(t)$.

$$V(t) = 12 \cos(\omega t + 20^\circ) = 12 \angle 20^\circ$$

$$i(t) = \frac{12 \angle 20^\circ}{2.4 \angle -45^\circ} = \frac{12}{2.4} \angle 20^\circ - (-45^\circ) = 5 \angle 65^\circ$$

$$i(t) = 5 \angle 65^\circ$$

$$i(t) = 5 \cos(\omega t + 65^\circ)$$

$$Z = -1.6971 + j1.6971$$

$$A = A e^{j\phi} = A \cos \phi + j A \sin \phi$$

$$|A| = \sqrt{R_c^2 + I_m^2} = \sqrt{(-1.6971)^2 + (1.6971)^2} = 2.4$$

$$Z = -1.6971 + j1.6971 \Rightarrow 2.4 \angle -45^\circ$$

$$\Delta \phi = \tan^{-1}\left(\frac{I_m}{R_c}\right) = \tan^{-1}\left(\frac{1.6971}{-1.6971}\right) = -45^\circ$$

b. (1 point) Taking the voltage $v(t)$ and the calculated current $i(t)$ from part 'a' of this problem, compute the real power P , reactive power Q , and apparent power $|S|$.

Real Power "P"

$$P = \frac{V_m I_m}{2} \cos(\phi_v - \phi_i) = \frac{12 \cdot 5}{2} \cos(20^\circ - 65^\circ) = 21.21 \text{ W}$$

Reactive Power "Q"

$$Q = \frac{V_m I_m}{2} \sin(\phi_v - \phi_i) = \frac{12 \cdot 5}{2} \sin(20^\circ - 65^\circ) = -21.21 \text{ VAR}$$

Apparent Power "|S|"

$$|S| = \sqrt{P^2 + Q^2} = \sqrt{(21.21)^2 + (-21.21)^2} = 29.99 \text{ VA}$$